

BOOKHUB SECOND-HAND BOOK MARKET PLACE Report

**DIPLOMA IN COMPUTER SCIENCE WITH
ARTIFICIAL INTELLIGENCE**

**WEB DEVELOPMENT
GROUP PROJECT**

**Batch: 25.1
Group Members: 028, 029, And 048**

CONTENTS

1. Introduction

- 1.1 Background
- 1.2 Problem Statement
- 1.3 Purpose
- 1.4 Scope

2. Project Objectives

- 2.1 Main Objectives
- 2.2 Specific Objectives

3. Requirements Analysis

- 3.1 Functional Requirements
- 3.2 Non-Functional Requirements
- 3.3 Hardware and Software Requirements

4. System Design

- 4.1 System Architecture
- 4.2 Database Design
- 4.3 User Interface Design

5. Implementation

- 5.1 Overall System Architecture
 - 5.1.1 Description of how the system works
 - 5.1.2 Frontend and backend overview

5.2 Fronted System

5.2.1 Description of Page Layout

5.2.2 Color Scheme and Fonts

6. Technology

6.1 HTML

6.2 CSS

6.3 JavaScript

6.4 Frameworks

6.5 Hosting platform

7. System Implementation

7.1 Description of Major Files

7.2 Explanation of important features

7.3 Screenshots.

8. Challenges and Solutions

8.1 Layout alignment issues

8.2 Responsive design problem

8.3 JavaScript errors

9. Future Enhancements

9.1 Additional features

9.2 Improvements planned

9.3 Scalability ideas

10. Conclusion

11. References

1. INTRODUCTION

1.1. Background of the project

BookHub is a web application that manages and displays books online. It allows users to easily view book information through a simple, user-friendly interface. The system organizes book details in a digital format, making access faster and more convenient.

1.2 Problem statement

Managing books manually is not efficient and takes a lot of time. It is hard to keep records current, find specific books, and prevent mistakes when relying on paper or manual systems.

1.3 Purpose of the project

The purpose of BookHub is to offer a simple and dependable platform for viewing and managing books online. The system allows users to access book information quickly. It also cuts down on manual work and improves accuracy.

1.4 Scope of the project

The scope of the proposed project includes both the frontend and basic backend functionality. The basic functionality includes the display of books, simple data management, and user interactions, and it does not encompass advanced functionality such as payment systems or analytics.

2. OBJECTIVE

2.1 Main Objective

The main objective of this project is to design a website with the ability to organize and show books in an effective and hand way.

2.2 Specific Objective

- Display book
 - It makes it possible for the user to display the books clearly.
- Admin 3Dashboard
 - The admin dashboard facilitates easy management of books and system content.
- Functional User Interface
 - The site is quite functional on a computer, tablet, and smartphone.

3. Requirements Analysis

3.1 Functional Requirements

- The user can view book information like title, image, and description.
- The user can view the list of available books.
- The system allows registration of users and login.
- The user can upload books to be sold.
- The user can have control over the books they upload.
- The admin can log in to the admin dashboard.
- The admin can add, modify, and delete book entries.
- The admin can manage system content.

3.2 Non-Functional Requirements

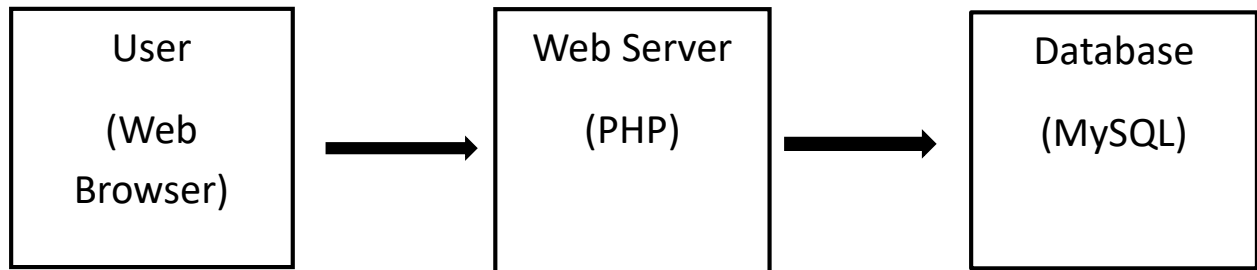
- This system should be easy to use and should be user-friendly.
- It should be responsive on various screen sizes.
- The system should allow secure log-in via sessions.
- The system should load pages quickly.
- The system should function properly on common web browsers.

3. Hardware and Software Requirements

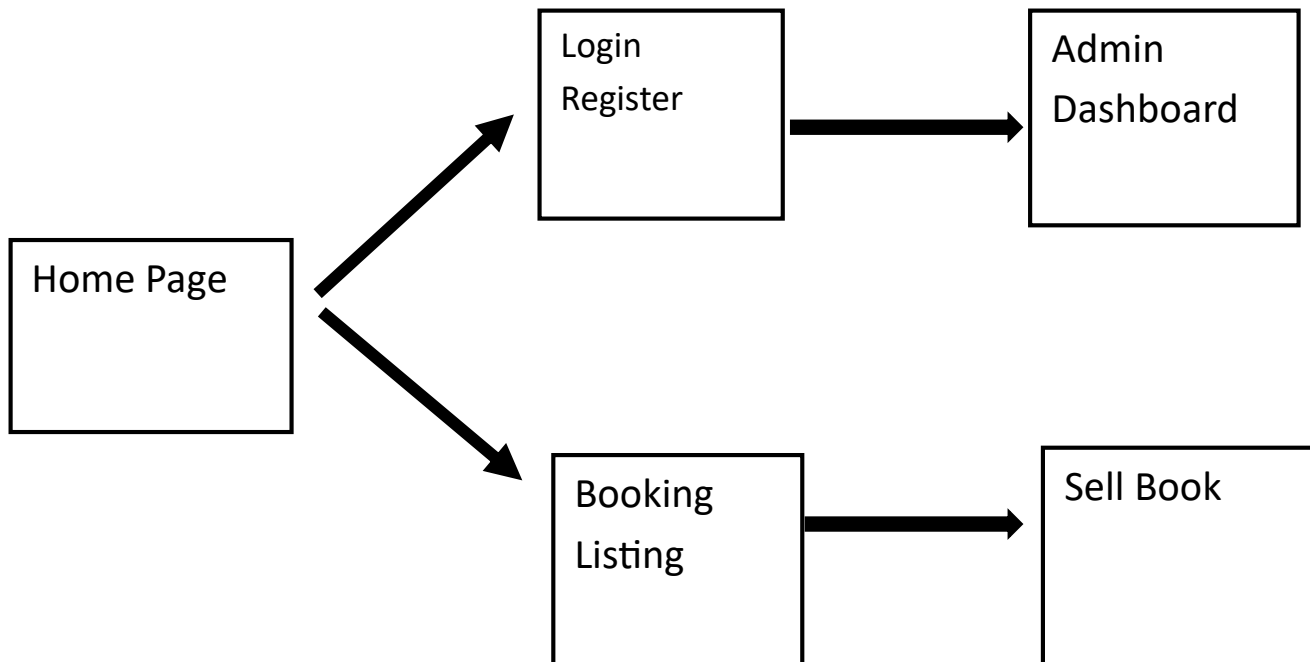
- Hardware Requirements
 - Laptop.
 - 4 GM RAM.
 - Keyboard and mouse.
 - Internet connection
- Software Requirements
 - Operating System: Windows.
 - Web Browser: Google Chrome.
 - Code Editor: Visual Studio Code.
 - Web Server: XAMPP (Apache).
 - Database: MYSQL.
 - Programming Language: HTML, CSS, JavaScript, and PHP.

4. System Design

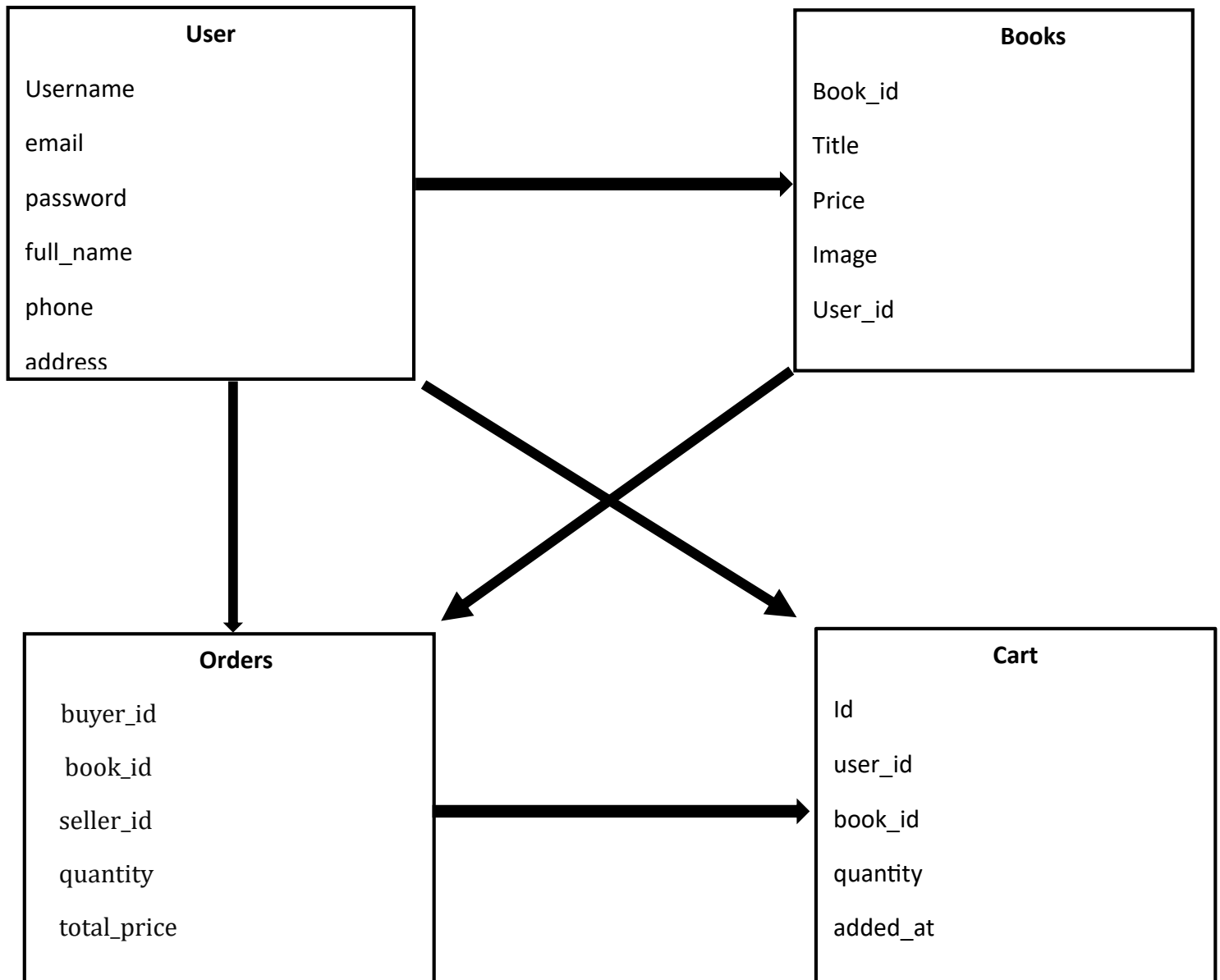
4.1 System Architecture



4.3 User Interface



4.2 Database Design



5. Implementation

5.1 Overall System Architecture

5.1.1 Description of How the System Works

When a user opens the BookHub website with a browser, requests like the display of books, logging in, uploading book information, and so on are generated. These requests are received by the web server. The requests are then processed by scripts written in PHP. Retrieval of information from and storage of data in the MySQL database take place as appropriate.

5.1.2 Fronted and Backend

Fronted:

The frontend part of the application relies on HTML, CSS, and basic JavaScript. It deals with displaying information, retrieving user data, and creating an attractive interface.

Backend:

Backend development is done using PHP and MySQL. PHP provides functionality related to authentication, book operations, and related databases, while all the data of the system is stored in the MySQL databases.

5.2 Fronted System

5.2.1 Description of page Layout

The layout of the webpage is made up of header, main content region, and footer region. The header region is where the title of the webpage and the navigation menu appear. The main content region shows books, forms, dashboards. Additionally, the basic information is located in the footer region.

5.2.2 Color Scheme and Font

The BookHub System has an effective and professional combination that promotes readability. The light colors are mixed with dark texts to enhance readability. The BookHub System employs web-safe fonts to ensure that the platform remains consistent regardless of the web browser used.

6. Technology

6.1 HTML

HTML creates the basic structure of the BookHub web pages. It defines page elements like headings, forms, buttons, images, and links. All user interface, including logging pages, book listings, and dashboards, are built with HTML.

6.2 CSS

CSS styles the web pages and improves their appearance. It controls layout design, colors, fonts, spacing, and responsiveness. CSS makes the BookHub interface visually clean, consistent, and user-friendly across different devices.

6.3 JavaScript

JavaScript adds basic interactivity and client-side validation to the system. It validates user input in forms and enhances user experience by providing immediate feedback without reloading pages.

6.4 Frameworks

This project does not use any major frontend or backend frameworks. The system is developed with core PHP for backend processing and plain HTML, CSS, and JavaScript for the frontend. This method promotes a better understanding of fundamental web development concepts.

6.5 Hosting Platform

The BookHub system is hosted locally using XAMPP, which provides an Apache web server and MySQL database. This platform is for development and testing purposes. The project can be deployed to a live hosting server in the future.

7. System Implementation

7.1 Description of Major Files

The BookHub project consists of several important files that handle different system functions:

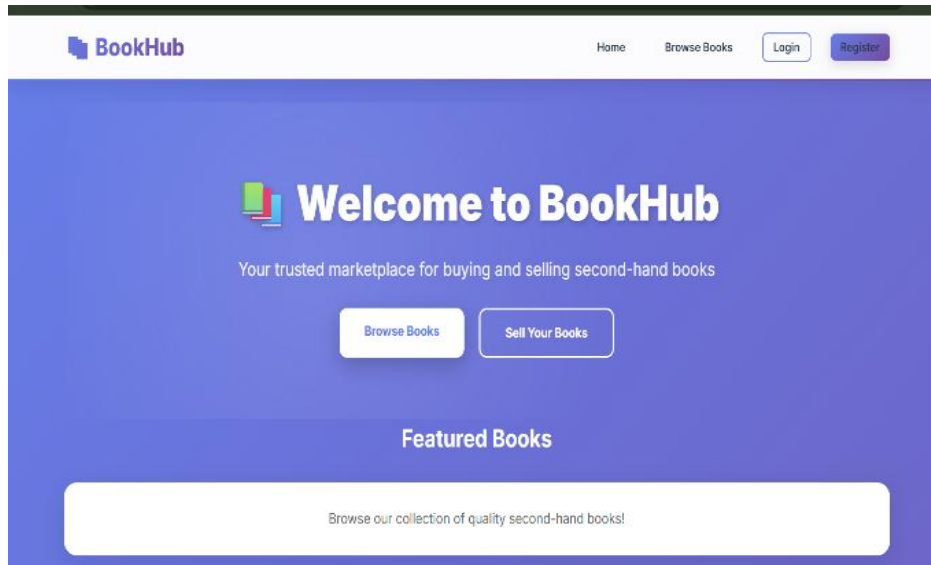
- index.php - Displays the home page and provides access to main navigation options.
- login.php - Handles user login functionality.
- register.php - Manages user registration and validation.
- book_list.php – Displays the list of available books.
- sell_book.php – Allows users to upload books for sale
- admin/dashboard.php – provides the admin dashboard for system management.
- config.php – Manages database connection settings.
- style.css – Contains styling rules for the user interface.
- Script.css – Handles basic JavaScript validation.

7.2 Explanation of important features

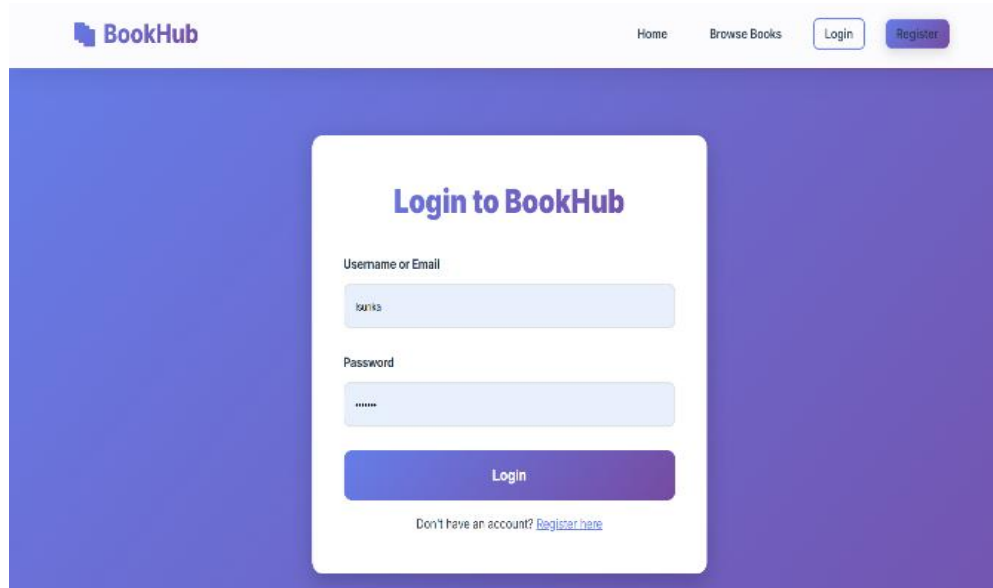
- **User Authentication:**
Enables secure login functionality with PHP sessions.
- **Book Display:**
Facilitates users to see the available books with details such as book title, price, etc.
- **Book Upload:**
The Users can upload book information and pictures to sell their books on the website.
- **Admin dashboard:**
Offers administrators facilitates to administer the books and system contents.
- **Database integration:**
Stores and fetches all data securely using MySQL Database.

7.2 Screenshots

- Home page

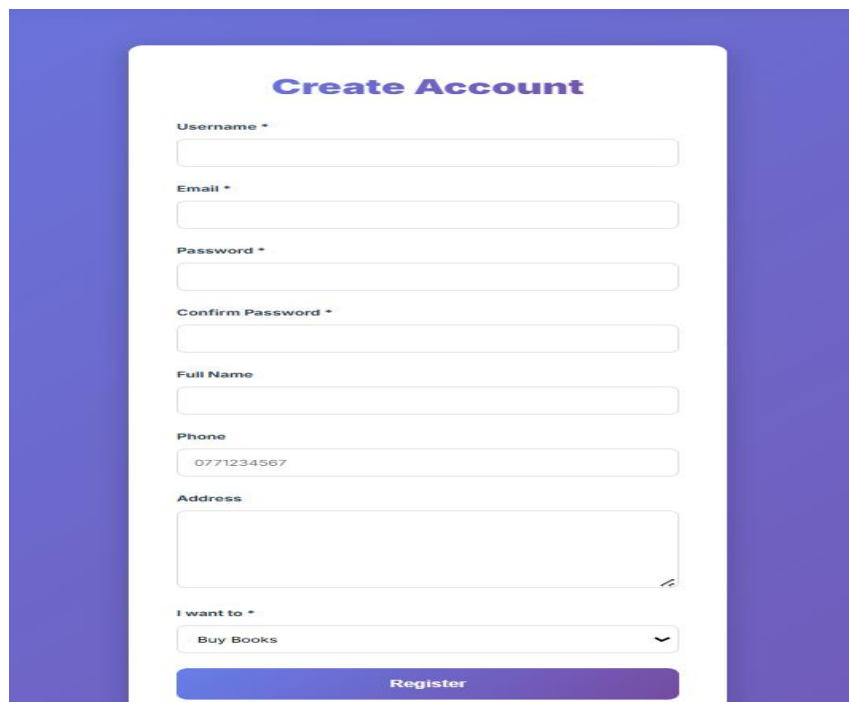


- Login Page



The screenshot shows the 'Login to BookHub' page. At the top, there is a navigation bar with the 'BookHub' logo, a 'Home' link, a 'Browse Books' link, and buttons for 'Login' and 'Register'. The main content area has a purple gradient background. In the center, a white card contains the title 'Login to BookHub'. Below the title are two input fields: 'Username or Email' with the text 'bunika' and 'Password' with masked characters. A blue 'Login' button is positioned below these fields. At the bottom of the card, there is a link that says 'Don't have an account? [Register here](#)'.

- Registration page



The screenshot shows the 'Create Account' page. The title 'Create Account' is at the top of the form. Below it are several input fields: 'Username *', 'Email *', 'Password *', 'Confirm Password *', 'Full Name', 'Phone' (with the number '0771234567'), and 'Address'. At the bottom of the form, there is a dropdown menu labeled 'I want to *' with the option 'Buy Books' selected. A blue 'Register' button is located at the very bottom of the form.

● Selling page

Sell Your Book

Book Title

Author

ISBN (International Standard Book Number)

Category

Select Category
▼

Condition

New
▼

Description

Describe the book's condition, any highlights, etc.

Selling Price (Rs.)

Original Price (Rs.)

Quantity Available

Book Cover Image

Choose File
No file chosen

Accepted formats: JPG, PNG (Max 5MB)

List Book for Sale

[View My Books](#)

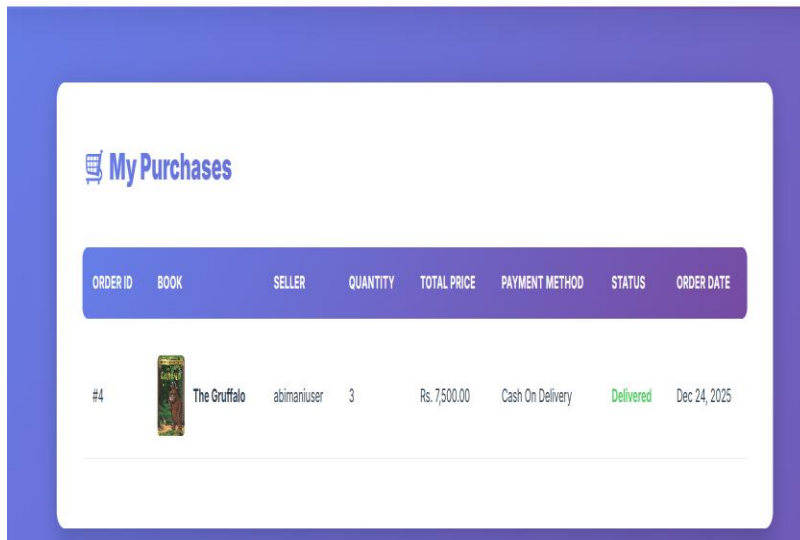
● My listed books

My Listed Books

+ Add New Book

Image	Title	Category	Price	Stock	Status	Views	Actions
	How to Teach Business English by Evan Frendo	Business	Rs. 3,900.00	✓ 2	Approved	0	View Delete
	Other Publications Information Technology CAMBRIDGE ICT STARTERS - NEXT STEPS STAGE 2 by Victoria Ellis, Sarah Lawrey, Doug Dickinson	Technology	Rs. 4,900.00	✓ 9	Approved	0	View Delete

- My purchase



The screenshot shows a web interface titled "My Purchases" with a shopping cart icon. Below the title is a table with the following columns: ORDER ID, BOOK, SELLER, QUANTITY, TOTAL PRICE, PAYMENT METHOD, STATUS, and ORDER DATE. A single order is listed with the following details:

ORDER ID	BOOK	SELLER	QUANTITY	TOTAL PRICE	PAYMENT METHOD	STATUS	ORDER DATE
#4	 The Gruffalo	abimaniuser	3	Rs. 7,500.00	Cash On Delivery	Delivered	Dec 24, 2025

8. Challenges and Solutions

8.1 Layout Alignment Issues

Challenge:

Some elements on the pages, for example, book cards, forms, and buttons, had not been aligned in a way that would ensure equal spacing on pages associated with different functions.

Solution:

CSS layout systems such as Flexbox were also applied for effective alignment of the elements. The margins and padding properties were adjusted accordingly, and universal CSS styles helped create uniformity on all pages.

8.2 Responsive Design Issues

Challenge:

The layout of this website was not displaying properly on screens that are smaller in size, like cell phones and tablets. Certain content was overlapping, and navigation links on small screens were difficult to operate.

Solution:

Techniques of responsive design were implemented using CSS media queries. Flexible units, percentage-based designs, and responsive images were used to ensure smooth functionality on desktops, tablets, and mobile devices.

8.3 JavaScript Errors

Challenge:

JavaScript validation scripts failed mostly, like in the form submission of login and registration, and book upload forms.

Solution:

The browser developer console was used to identify the bugs. Syntax errors, variable name verification, and proper form input validation at the JavaScript code level were done before submission.

9. Future Enhancements

9.1 Additional Features

Furthermore, in the future, some extra functions may be added, for example, integrating payment features in the web payment system, searching capabilities for books, and handling user profiles. These functions will enable the system to be more interactive and user-friendly.

9.2 Planned Enhancements

The following are improvements that will be worked on: enhancements in the user interface design, improving form validation, and strengthening security mechanisms like password encryption. Other enhancements can also be made for optimization to reduce page loading time.

9.3 Scalability

The scalability of the system can be done by hosting it on a hosting server and optimizing it for handling large amounts of data. The addition of support for increased numbers of users, books, and additional modules, such as mobile applications, can also be done

10. Conclusion

The BookHub project has been successfully developed as a web application, which can be utilized to manage and display books online. The application offers a very convenient platform to the user in displaying books, as well as to the admin in efficiently managing the information of the books. The use of concepts like HTML, CSS, JavaScript, PHP, and MySQL in the project applies the concepts of web development.

All the prime tasks that had to be accomplished in the project have been accomplished, such as book display, admin management, and design of the user interface. The project satisfies all the basic requirements that had to be met at the initial stages of the project. BookHub is a very effective solution for simple book management tasks, which can evolve further to include advanced features.

11. References

- **PHP official documentation:**
 - PHP Manual – server-side scripting and session
 - <https://www.php.net/docs.php>
- **MySQL Documentation:**
 - MySQL Reference Manual – Database design and queries
 - <https://dev.mysql.com/doc/>
- **MDN Web Docs:**
 - HTML, CSS, and JavaScript reference
 - <https://developer.mozilla.org/>
- **W3Schools:**
 - Tutorials for HTML, CSS, JavaScript, PHP, and MySQL
 - <https://www.w3schools.com/>
- **Apache Friends (XAMPP):**
 - Local hosting using Apache and MySQL
 - <https://www.apachefriends.org/>

- **phpMyAdmin Documentation:**
 - Database management tool used with MySQL
 - <https://www.phpmyadmin.net/docs/>
- **OWASP Web Security Basics:**
 - Web application security best practices
 - <https://owasp.org/>

Link of the website:

<http://bookmarket2025.gt.tc>

Website login admin panel

User name: admin

Password: admin123